DEVANSH AGRAWAL

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Education

University of Illinois Urbana-Champaign — Jan 2026 - May 2027 (Expected)
Master of Computer Science — Specializing in ML Systems · GPA: 4.00/4.00

Purdue University — Aug 2018 - May 2022
Bachelor of Science in Computer Science — Minor in Mathematics · GPA: 3.80/4.00

Work Experience

PEOPLELENS | AI Engineer Intern — Cupertino, CA | Jun 2026 – Aug 2026

- **Designed an LLM pipeline that scores sales calls on 8 coaching criteria, chunking transcripts and using hybrid retrieval (BM25 + dense embeddings) to ground every score in verbatim quotes**
- **Achieved 1.37 mean-absolute-error vs human experts on a 1–10 rubric (62% of scores within 1 point) with a two-stage analyze-then-judge LLM design gated by deterministic, code-based validation**
- **Co-developed the coaching-plan generator that turns scores into specific, next-step "nudges" for reps and their managers**
- **Shipped a multi-tenant MCP (Model Context Protocol) server exposing coaching data as 11 read-only tools to AI assistants in Slack, Teams, and Claude on GCP + MongoDB**
- **Verified per-company data isolation and scaled the multi-tenant serving path with a Kafka queue, rate limiting, and idempotent retries, load-tested under concurrent production traffic**
- **Cut LLM inference cost and latency via prompt caching, single-pass embedding, and bounded concurrency, reusing one transcript across all 8 per-rubric judges**

OPEXUS (FORMERLY AINS LLC) | Full Stack Software Engineer — Washington, DC | Jul 2022 – Nov 2025

- **Fine-tuned DistilBERT and DeBERTa-v3 (HuggingFace) on 8,000+ annotated documents, lifting redaction accuracy 80%+ over the CoreNLP baseline on ambiguous cases**
- **Engineered a multi-stage NLP redaction pipeline (CoreNLP + AWS Comprehend), cutting manual document review from 10 hours to 30 minutes per case**
- **Migrated a monolithic .NET 8 platform (300 agencies, 3.4M requests/yr) to Dockerized microservices, cutting deployment time 70% and enabling independent scaling of the AI service**
- **Led a team of 8 to ship bilingual French/Spanish support, winning 45 Canadian customers and growing revenue 15%**
- **Automated a CI/CD security pipeline (Fortify, Qualys) resolving 1,500+ vulnerabilities; integrated Relativity, Pay.gov, and NS Pay APIs via SAML and MFA**

PURDUE UNIVERSITY | Applications Developer — Fort Wayne, IN | May 2021 – May 2022

- **Developed internal applications in Java, JavaScript, and React; automated SQL tasks via triggers and scheduled jobs, cutting DBA workload 15% and reducing support tickets 15%**

Research

LIVER CT RECONSTRUCTION USING MACHINE LEARNING Purdue University

- **Preprocessed liver CT datasets in Python (ITK, OpenCV, NumPy) with 2D/3D segmentation, affine/deformable registration, and volume normalization**
- **Trained a U-Net pixel-wise regression model to synthesize multi-angle CT projections (MSE, MAE, SSIM, PSNR), enabling 25–50% radiation dose reduction**

NEURODRONE: AUTONOMOUS NAVIGATION VIA DEEP RL Purdue University

- **Benchmarked 4 recurrent RL agents (RDQN, RDDPG-PER, TD3-PER, RA2C) for drone obstacle avoidance via occupancy mapping**

Skills

ML & Deep Learning: **PyTorch, HuggingFace Transformers, fine-tuning, transfer learning, NER, U-Net, ONNX, scikit-learn, Pandas, NumPy, OpenCV, ITK**

LLMs & GenAI: **RAG, BM25 + dense/semantic retrieval, embeddings (bge-m3), vector search, LLM-as-a-judge, prompt engineering, prompt caching, grounding, hallucination mitigation, LLM evaluation, LLM serving (vLLM), Anthropic (Claude), Gemini, AWS Comprehend, Stanford CoreNLP**

Backend & Distributed Systems: **MCP/FastMCP, REST APIs, multi-tenant architecture, tenant isolation, JWT, OAuth 2.0, SAML, MFA, rate limiting, idempotency, Kafka message queues, microservices, .NET, Node.js, React**

Cloud, Data & DevOps: **GCP (Cloud Run, Firestore), AWS, Docker, Azure DevOps, CI/CD, Git, MongoDB (Atlas), MSSQL, MySQL, Slack API, Microsoft Bot Framework, Fortify, Qualys**

Languages: **Python, C#, C++, Java, JavaScript, SQL, Bash**